

# Shakiba Ahmadimehr

BIOMEDICAL ENGINEER | NEUROSCIENCE RESEARCHER  
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## Research Interest

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- **Biomedical image/signal processing**  
Task-based / Resting-state Brain mapping  
Neuroimaging using MRI, DTI, fMRI, EEG  
Advance Signal processing
- **Computational Neuroscience**
- **Biomedical Pattern Recognition**
- **Machine Learning with the medical approach**
- **Cognitive Neuroscience**  
Studying Human Brain with its network such as Cognition, Language, Learning & Memory  
Studying neurologic & neuropsychiatric disorders

## Education

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**University of California, Irvine**  
**PhD Student at department of Biomedical Engineering**

Aug 2023- Jun 2025  
*Orange County, CA, USA*

**Lab rotation Project 1:** Quantifying Brain Structural Connectivity Using Diffusion Tensor Imaging (DTI)  
Under the supervision of Prof. Terence David Sanger

**Lab rotation Project 2 & 3:** Decoding Neural and Behavioral Impacts of Attentional Cues Using fMRI and Machine Learning.  
Under the supervision of Prof. Emily Grossman

**Tulane University**  
**Research Assistant of Bioengineering and Biomedical Engineering**  
**Project Title:** Extraction and integration of biomarkers from multi-modality MRI using deep learning approaches.  
Under the supervision of Prof. Yu-ping Wang

Aug 2022- May 2023  
*New Orleans, LA, USA*

**Master of Science in Biomedical Engineering**  
South Tehran Branch, Islamic Azad University (IAUSTB)  
College of Engineering  
Cumulative GPA: 3.76 / 4.0 (18.33 / 20) (Ranked student)  
**Thesis Title:** Evaluation of brain connectivity using fMRI data in Parkinson's disease  
Under the supervision of Dr. Soroosh Behbahani

Sep 2019- Jan 2022  
*Tehran, Iran*

**Bachelor of Engineering in Biomedical Engineering**  
Tehran Medical Branch, Islamic Azad University (IAUTMU)  
College of Health and Biomedical Engineering  
Cumulative GPA: 3.65 / 4.0 (17.80 / 20) (Ranked student)  
**Senior Project Title:** Implementation of Machine learning methods to discover attractive and unattractive faces  
Under the supervision of Dr. Mohammad Karimi Moridani

Sep 2015- Feb 2019  
*Tehran, Iran*

**DIPLOMA IN MATHEMATICS AND PHYSICS**  
**DISCIPLINE**

Sep 2011 - Aug 2015  
*Tehran, Iran*

Farzanegan high school, National Organization for Development of Exceptional Talents, are national selective School in Iran developed specifically for the development of exceptionally talented Students  
Cumulative GPA: 3.70 / 4.0 (18.52 / 20) (Ranked student)

## Research Experience

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- Investigating the functional and effective connectivity by combining ICA and GCA as well as transfer entropy in Parkinson's disease (PD) patients using fMRI image data at rest.
- Investigating Kernel Mutual Information on Resting-State networks' regions via graph theory
- Evaluating Facial Attractiveness through Proportions Analysis Based on Geometric Features.
- Implementation of geometric features to discover attractive and unattractive faces using a machine learning approach
- Cell detection with the application of Biomedical Image processing and machine learning method.
- Evaluation of EEG and fMRI signals of the brain between psychiatric patients and healthy individuals and extract mathematical features
- Prediction of the trend of increasing COVID-19 statistically and mathematically
- Structural Connectivity Analysis in the Brain Using DTI
- Investigating the Role of Attentional Cues in Visual Discrimination: A Machine Learning and fMRI Approach

## Teaching & Work Experiences

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### Teaching Assistant, Graduate Level Course- Biological Systems Modeling and Biological Signal Processing

Department of Electric and Electronic Engineering, South Tehran Branch

### Teaching Assistant, Graduate Level Course- Differential Equations and Engineering math

Department of Biomedical Engineering, Tehran Medical Branch

- Conducted office hours to help students understand and solve homework problems
- Prepared and graded homework solutions
- Wrote weekly quizzes, posted solutions online, graded quizzes and exams, and kept a record of the scores using Excel
- Collaborated with professors and other TAs on course material and grading policies, improving communication skills

## Honors & Achievements

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- Fully funded Ph.D. admission in Biomedical Engineering at UC, Irvine, CA Feb 2023  
*Orange County, CA, USA*
- Fully funded Ph.D. Research Assistant offer in Bioengineering and Biomedical Engineering at Tulane University, MBB Lab March 2022  
*New Orleans, LA, USA*
- One of the Top 10 students among other 200 students in the Biomedical Engineering program at IAUTMU Feb 2019  
*Tehran, Iran*
- One of the Top 3 students among other 70 students in the Biomedical Engineering program at IAUSTB Jun 2020  
*Tehran, Iran*

## Publications

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### Journal publications:

- **Ahmadimehr, Shakiba**, and Mohammad Karimi Moridani. "Evaluating Facial Attractiveness through Proportions Analysis Based on Geometric Features." *Journal of Image Processing & Pattern Recognition Progress* 7.2 (2020): 20-26.
- **Ahmadimehr, Shakiba**, and Mohammad Karimi Moridani. "Identify Attractive and Unattractive Individuals Based on Geometric Features Using Neural Network." *Traitement du Signal* 38.4 (2021): 1007-1012.
- **Ahmadimehr, Shakiba**, "Alteration of the resting-state brain networks: a multi-aspect study of functional and effective connectivity in Parkinson's disease.", [Preprint].

- **Ahmadimehr, Shakiba**, " Investigation of Linear and Non-linear functional connectivity within resting-state networks using graph theory in Parkinson's disease.", [Preprint].
- Anton Orlichenko, **Shakiba Ahmadimehr**, Gemeng Zhang, Gang Qu, Zhengming Ding, Yu-Ping Wang, Dynamic dictionary entries are rank-1 functional connectivity networks associated with maturation [Presented at OHBM Conference – July 2023]

## Skills

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- **Programming** Matlab, Python
- **Software** FSL, MATLAB, MRICron, SPM, CONN, BCT, BrainNET Viewer, EEGLAB, SPSS, AFNI, FreeSurfer,
- **OS** Linux, Windows
- **Languages** Persian (Native), English (TOEFL, speaking: 22, Listening:29, Reading:30, Writing: 28), German (A1)

## References

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### Prof. Terence David Sanger

[Electrical Engineering and Computer Science](#)

Vice President, Chief Scientific Officer

[CHOC](#)

Vice Chair of Research for Pediatrics

[School of Medicine](#)

Staff Physician

[CHOC Department of Neurology](#)

Email: [tsanger@uci.edu](mailto:tsanger@uci.edu)

### Prof. Emily Grossman

Professor, Cognitive Sciences

School of Social Sciences

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### Prof. Yu-Ping Wang

Department of Biomedical Engineering,

School of Science & Engineering, Tulane University

Phone: 1.504.865-5897

New Orleans, LA 70118-5698

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### Dr. Soroor Behbehani

Assistant Professor,

Department of Electric and Electronic Engineering,

Islamic Azad University, South Tehran Branch

Email: [Sor.behbahani@gmail.com](mailto:Sor.behbahani@gmail.com)

Relationship: MSc. Thesis Supervisor

### Dr. Mohammad Karimi Moridani

Assistant Professor,

Department of Biomedical Engineering,

Islamic Azad University, Tehran Medical Sciences,

Email: [Karimi.m@iautmu.ac.ir](mailto:Karimi.m@iautmu.ac.ir)

Relationship: B.Sc. project Supervisor